

MI-08: Partial Result

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Repository status: Partially resolved. The complete argument received an independent Codex-agent PASS review for the stated partial result only. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope.](#)

1 MI-08: fixed and adaptive orthogonal pinching have the same length

This section is partial progress on MI-08, not a determination of the length in every dimension. It resolves the comparison between the fixed and input-dependent versions in [1, Questions 4.8–4.9] and determines additional exact dimensions.

Let $\Delta(X) = \text{diag}(x_{11}, \dots, x_{dd})$ for real $d \times d$ matrices. Let $f(d)$ be the least q for which there are fixed orthogonal matrices $O_1, \dots, O_q$ satisfying

$$\Delta(X) = \frac{1}{q} \sum_{i=1}^q O_i X O_i^T \quad \text{for every real } X.$$

Let $a(d)$ be the least q for which the same identity holds for every real X , allowing the matrices O_i to depend on X . Finally, let

$$h(d) = \min\{q : \exists H \in \{-1, 1\}^{q \times d}, H^T H = qI_d\}.$$

Such a q exists, for example by the Sylvester construction of sign matrices of power-of-two order.

Theorem 1.1. *For every positive integer d , one has*

$$f(d) = a(d) = h(d).$$

In particular, $f(d) = a(d) = 12$ for $9 \leq d \leq 12$.

Proof. If H realizes $h(d)$, set $O_i = \text{diag}(h_{i1}, \dots, h_{id})$. Orthogonality of the columns of H gives

$$\frac{1}{q} \sum_i O_i X O_i^T = \Delta(X)$$

entry by entry, for every real X . Thus $a(d) \leq f(d) \leq h(d)$.

For the reverse inequality, choose $D = \text{diag}(1, 2, \dots, d)$ and the skew-symmetric matrix K with $k_{ij} = 1$ for $i < j$ and $k_{ij} = -1$ for $i > j$. Set $X = D + K$. Suppose q orthogonal matrices, even chosen specifically for this X , satisfy the required identity. Taking symmetric parts yields

$$\frac{1}{q} \sum_i O_i D O_i^T = D.$$

Every term has the same Frobenius norm as D . Therefore

$$\sum_i \|O_i D O_i^T - D\|_F^2 = 2q \|D\|_F^2 - 2 \operatorname{tr} \left(D \sum_i O_i D O_i^T \right) = 0.$$

Each O_i consequently commutes with D . Since the diagonal entries of D are distinct, O_i is a diagonal sign matrix. Taking skew-symmetric parts now shows that, for every $j \neq l$,

$$0 = \sum_i (O_i)_{jj} (O_i)_{ll} k_{jl}.$$

All k_{jl} are nonzero. The $q \times d$ matrix of these signs therefore has orthogonal columns, so $q \geq h(d)$. This proves $a(d) \geq h(d)$ and establishes the equality.

For $d \geq 3$, any admissible q is divisible by four. Indeed, multiply rows by signs to make the first column all ones. The second and third columns are both balanced and orthogonal; the four possible sign pairs must each occur $q/4$ times. Also $q \geq d$ by rank. Thus $h(d) \geq 12$ for $9 \leq d \leq 12$.

For completeness, an order-twelve sign matrix can be constructed entirely explicitly. In the field $\mathbb{F}_{11}$, put $\chi(0) = 0$, let $\chi(x) = 1$ for nonzero squares, and let $\chi(x) = -1$ otherwise. Let $Q_{ij} = \chi(i - j)$ for $i, j \in \mathbb{F}_{11}$. Then $Q^T = -Q$, $Q\mathbf{1} = 0$, and $QQ^T = 11I - J$. The last identity can also be checked directly using the square residues $\{1, 3, 4, 5, 9\}$; the accompanying verifier does so with integer arithmetic. Hence

$$H = \begin{pmatrix} 1 & \mathbf{1}^T \\ \mathbf{1} & Q - I \end{pmatrix} \quad \text{satisfies} \quad HH^T = 12I_{12}.$$

Taking any d columns gives $h(d) \leq 12$ for $d \leq 12$, proving the claimed values. $\square$

Remark 1.2 (What remains). The general value of $h(d)$ is not determined here. In particular, $h(4m) = 4m$ is equivalent to the existence of a Hadamard matrix of order $4m$. The value $h(9) = 12$ also rules out a universal dichotomy between d and the next power of two. The adaptive comparison above uses all real matrices, including nonsymmetric ones, exactly as in the stated question; it is not a claim about a restriction to symmetric inputs only.

References

- [1] Jean-Christophe Bourin and Eun-Young Lee. Averages over matrix unitary orbits and spectral order. arXiv:2606.15624, 2026. Version consulted: v2; Question 3.11, Conjecture 3.12, and Questions 4.8–4.9. URL: <https://arxiv.org/abs/2606.15624>.